Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Automated Calibration and High-Speed Precision Motion Programming for Multi-Robot Manufacturing Cells			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (**X**), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Control and Intelligence Technologies	Robotics/Automation Machinery	Robot Vision and Production Automation Technologies
Concept of Proposed Technology	We propose to develop software and procedures for rapid commissioning and high-throughput, high-accuracy operation of single- and multi-robot arm manufacturing cells. The core concept is to integrate: • automatic calibration for fast commissioning of single- or multi-arm robots in manufacturing, • configuration/pose optimization to maximize feasible speed while maintaining accuracy and avoiding singularities/collisions, and • motion primitives optimization that converts optimized trajectories into commands (e.g., linear/circular segments, blended moves) that are native to industrial robots to produce reliable high-speed high-accuracy motions on standard industrial robots. The system learns and compensates for controller- and cell-specific effects (e.g., blending, interpolation, dynamic limitations) through a combination of model-driven and data-driven optimization, producing motion programs that achieve uniform path speed, tight accuracy, and robust repeatability across long spatial curves and complex 3D paths.			

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	Potential uses include robotic coating/cold spray, arc/laser welding, grinding/polishing, additive manufacturing, and inspection, especially in SMEs where target products and processes can change frequently and commissioning time directly impacts cost and competitiveness.
Rationale for Joint R&D Project	Manufacturing SMEs increasingly deploy robotic technologies, but they face persistent barriers in extracting high value from robotic manufacturing cells [***Marvel2015***, ***Messina2024***], such as: • long commissioning time due to tedious calibration and trial-and-error tuning, • difficulty in reconfiguration for new products or processes, • limited throughput because high-speed motion often causes inaccurate motion, speed variations, and quality defects, and • complexity of multi-robot cells including coordination, calibration, and simultaneous motion control. This project is needed to close the gap between algorithms produced in academic setting at RPI (validated at TRL 3) and shop-floor deployable tools that meet SME constraints (limited engineering time, heterogeneous controllers, frequent changeovers). In this collaboration, RPI will contribute the foundational research in trajectory optimization, motion-primitive calculation, and system identification/learning for controller-aware refinement. The Korean SME, in collaboration with RPI, will define requirements and acceptance tests, provide the target manufacturing processes, representative parts, robot controllers and interface, and allocate engineering resources to integrate the software into their cell. Both parties will collaborate on running structured experiments to assess performance, usability, and robustness. The Korean SME will thus provide realistic parameters and deployment scenarios that cannot be captured in a university-only environment, which are necessary to develop the technology to TRL 6. Joint R&D ensures the technology is advanced, implementable, and aligned with commercialization paths within Korean industrial settings.
Key Words	1. Robotic manufacturing 2. Automated calibration in commissioning 3. Pose and configuration optimization 4. High-speed high-accuracy path tracking 5. Motion primitives optimization for industrial robots

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level 3	**End of the Project**	Level 6
R&D Objectives	1. Automatic calibration and fast commissioning for multi-robot cells: develop an automated calibration pipeline for robot-to-world, tool center point (TCP), and inter-robot/frame registration suitable for SMEs, including procedures for rapid re-calibration and drift monitoring. 2. Configuration/pose optimization for high-speed, high-accuracy execution: develop an optimization tool that selects robot configurations and end-effector poses along spatial curves to maximize feasible speed subject to accuracy, joint/velocity/acceleration/jerk limits, singularity avoidance, collision constraints, and tool-orientation constraints. 3. Motion-primitive optimization and controller-aware program synthesis: convert optimized continuous trajectories into industrial-robot-controller native programs using motion primitives (linear/circular/blended moves) with minimal waypoints while preserving accuracy and speed			

	uniformity. 4. Data-driven refinement to compensate controller-specific effects: implement iterative, sample-efficient correction of waypoint placement and blending parameters using measured deviations to achieve consistently high performance. 5. TRL 6 demonstration in representative SME processes: integrate the toolchain into a pilot cell (single-robot and multi-robot variants) and demonstrate repeatable production on representative parts with documented throughput and quality gains.
R&D Target Specifications	**Commissioning / Calibration** A1. Initial cell commissioning time: ≤ 4 hours from power-on to verified coordinate registration (robot base, TCP, workpiece frames). A2. Re-calibration time after changeover: ≤ 60 minutes. **High-Speed High-Accuracy Motion** B1. Absolute TCP positioning accuracy after calibration: ≤ 1 mm maximum tracking error and ≤ 0.5 mm RMS tracking error in the working volume. B2. Inter-robot frame registration error (multi-robot cells): ≤ 0.5 mm translation and ≤ 0.3° rotation. B3 Speed uniformity along process path: ≥ 90% of commanded speed maintained; speed ripple ≤ 10% (peak-to-peak/mean). **Efficiency and Robustness** C1. Cycle-time reduction vs. baseline SME program: ≥ 25% on at least two representative parts. C2. Repeatability across 8-hour production run: drift in contour error ≤ 20% of initial value. C3. System portability: support at least two industrial controller families (or two robot platforms) in the project period.
R&D Plan	This section presents the detailed research plan to advance the proposed technology from TRL 3 to TRL 6, organized as (1) supporting results, (2) technical approach, and (3) work plan, timeline, and milestones. **1. Supporting Results** Our team has already established a strong technical foundation for the proposed system, with validated results in automated robot calibration, high-speed/high-accuracy curve tracking, and optimization of motion-primitives for industrial robots. All these results will support the proposed research. **A. Configuration-dependent kinematics model and calibration** We developed a configuration-dependent calibration framework that explicitly models non-geometric effects (e.g., compliance, backlash) that vary with robot pose and loading. Our results significantly improve upon vendor-provided kinematics model that are static.

Key contributions in this direction include: (i) a calibration workflow based on optimization of Product-of-Exponentials (POE) model and (ii) interpolation across the workspace using Fourier basis that are parameterized by key joint angles (e.g., shoulder/elbow). The latter is an efficient and physically motivated alternative to purely data-driven approximators. See Fig. 1.

Experimental validation on two 6R industrial robots demonstrated that the method reduced maximum Tool Center Point (TCP) error by >50%, achieving ≤1 mm maximum position error in test sets. The calibrated models were further evaluated in a dual-robot collaborative contact test, supporting applicability to multi-robot cells.

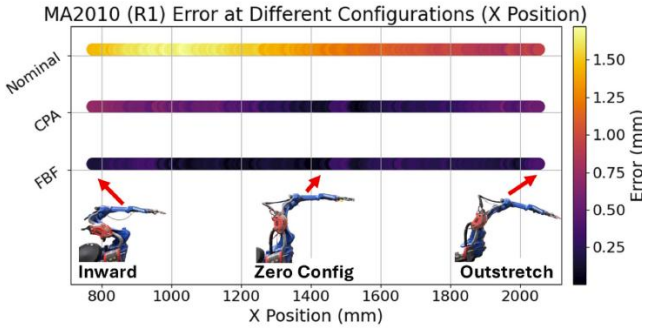

Fig 1. Results of automated calibration on Fanuc MA2010 robot. Bars indicate position error across configurations for three different calibration methods. The proposed method (FBF: Fourier Basis Functions) demonstrates the benefit of configuration-dependent calibration in maintaining low error throughout the workspace. See [***Lu2025***].

B. High-speed, high-accuracy spatial curve tracking using motion primitives for single-robot and multi-robot deployment

We developed a systematic optimization framework for industrial robots programmed via motion primitives (Cartesian linear/circular and joint primitives) where controller internals are not exposed. The framework explicitly addresses the real-world challenge that industrial robot motion depends on servo behavior, curve blending, and joint constraints that are not known to the users.

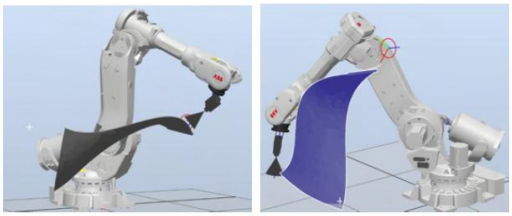

Fig 2. Simulated pose optimization in ABB RobotStudio. Left: baseline pose. Right: optimized pose. See [***He2023***] for details.

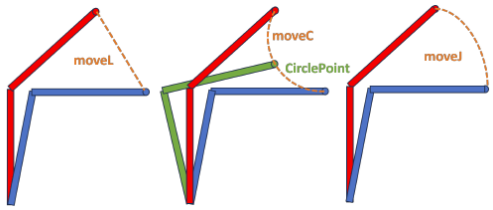

Fig 3. Typical motion primitives in industrial robots: *moveL*: linear in Cartesian Space; *moveC*: circular arc in Cartesian space; *moveJ*: linear in joint space. See [***He2024***].

Our approach optimizes (i) discrete parameters including redundant degrees of freedom and robot pose/configuration (see Fig. 2), (ii) the motion primitive parameters including segment types and curve blending parameters (see Fig. 3), and (iii) waypoints through iterative updates that directly reduce tracking error. We demonstrated our results on both ABB and FANUC robots (see Fig. 4), showing substantial performance gains relative to baseline industry practice, which includes more than 2x speed improvement in representative challenging spatial curves while enforcing tracking accuracy and speed-uniformity constraints.

We extended the motion-primitive optimization concept to dual-robot setups used to overcome single-robot speed limits and support multi-robot cooperative processes [***He2024***]. For this configuration, we developed a three-step design workflow that includes (i) selecting relative robot placement based on the specified relative path, (ii) selecting relative waypoints and motion primitives, and (iii) iteratively refining parameters/waypoints to meet accuracy and uniformity constraints at maximum speed. In a realistic simulation (FANUC RoboGuide), this workflow achieved very large throughput gains over baseline programming practice, with ~6x speed improvement in representative challenging spatial curves while maintaining bounded tracking error and improving speed uniformity and significant improvements over already optimized single-robot operation.

Fig 4. Industrial robot experimental platforms (ABB robots) used for validating high-speed, high-accuracy motion-primitive execution for complex paths on curved surfaces [***He2024***].

C. Motion profile optimization/refinement using reinforcement learning

We also developed a bi-level strategy that optimizes execution accuracy using reinforcement learning, with policy learning in a vendor-specific realistic simulation (ABB RobotStudio) and optimizes execution speed using bisection search for the fastest motion that still meets accuracy/uniformity specifications [***Wen2023***]. Experiments (see Fig. 5) show that the learned policy reduces the optimization cost required to meet desired specifications and can generalize to trajectories not included in training, supporting sample-efficient refinement under controller uncertainty.

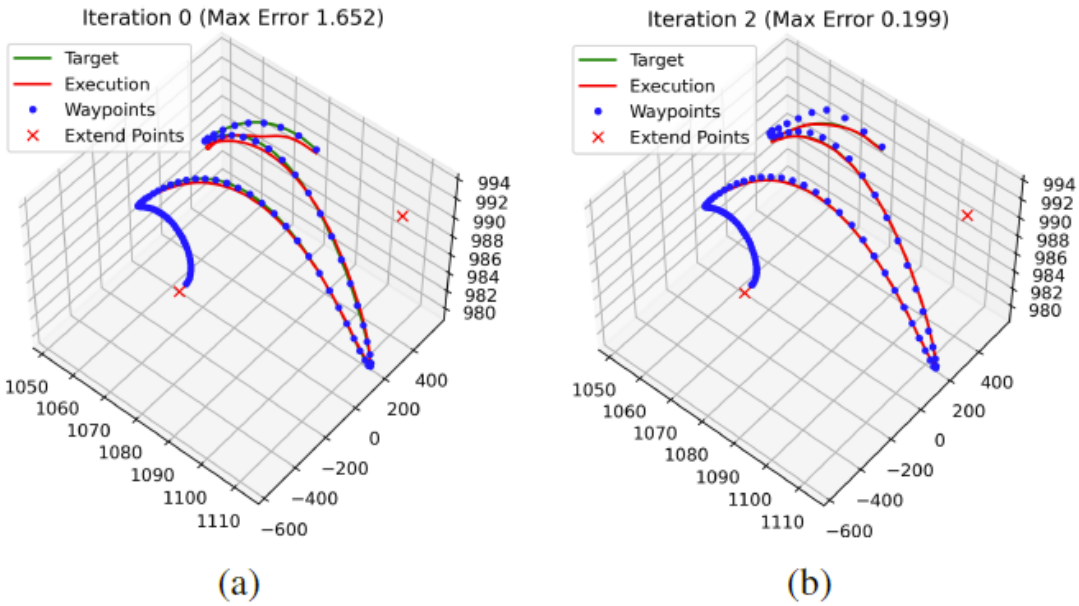

Fig 5. Representative TCP trajectories showing baseline execution (a) versus reinforcement learning-based motion profile optimization (b), where tracking error (measured in mm) is reduced to a sub-mm precision while preserving speed objectives. See [***Wen2023***].

Collectively, these results demonstrate that the technical approach already works on representative robots and trajectories, and that we have early evidence of portability across robot families. Thus, we have a clear pathway to SME-grade commissioning and operation.

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2. Technical Approach

In this project, we will translate existing results presented in the previous section into an SME-deployable system through a joint research project that couples (i) process definition and system specification, (ii) system/algorithm development, and (iii) repeated integration and validation on the SME pilot cell.

The technical approach is organized as the following workflow: define → baseline → build → integrate → test → refine → pilot, which is explained below.

A. Process definition and system requirements and specifications

Korean SME and RPI will jointly document the target manufacturing process(es) (e.g., welding, coating/cold spray, polishing), including: part geometry and path definition sources (CAD/scan/teach), process constraints (tool orientation ranges, working distance limits, keep-out zones), quality criteria (e.g., required accuracy), and production constraints (commissioning time and cycle-time targets, changeover frequency). This information will be translated into a measurable acceptance test suite aligned with the project's target specifications (commissioning time, contour error, speed uniformity, repeatability).

B. Baseline benchmarking on Korean SME manufacturing cell

The team will establish a performance baseline using the SME's current programming practice (manual waypoint definition and tuning). We will measure specifications outlined in part A, e.g., commissioning time, cycle time, path accuracy, and speed variation, and then perform an error analysis to identify dominant contributors (calibration/registration error, configuration-dependent effects, primitive/blending effects, dynamic limits). This baseline serves two purposes: it provides quantitative reference for improvement claims and it identifies which system modules (calibration, pose optimization, primitive optimization, refinement) should be prioritized for the project.

C. Development of core technical modules

We will develop and package the system as four integrated modules that address SME needs and are compatible with the built-in controller of industrial robots.

1. Automatic calibration and fast commissioning: procedures and software for base/workpiece/TCP calibration and (when needed) multi-robot calibration, with verification checks and fast re-calibration after changeover.
2. Pose/configuration optimization: compute configuration/pose schedules that maximize feasible speed while maintaining accuracy, avoiding singularities/collisions, and respecting process constraints (e.g., tool orientation and standoff distance).
3. Motion-primitive program synthesis and optimization: translate optimized plans into controller-native motion primitives (linear/circular/joint moves) with minimal waypoints, tuned blending, and speed commands suitable for reliable high-speed execution.
4. Measurement-driven refinement: use on-cell measurements to update waypoints and blending parameters iteratively in order to compensate controller- and cell-specific behavior with low number of trials.

D. Iterative implementation, integration, and testing on the SME pilot cell

We will run multiple integration and testing rounds on the SME manufacturing cell. In each round, the team will deploy the current system build, execute a standardized test set on representative parts/paths, and collect performance metrics (accuracy, speed uniformity, cycle time, commissioning time). Each iteration will lead to updated algorithms and procedures based on observed gaps. This loop is essential for achieving TRL 6 because it exposes controller-specific blending behavior, process-specific constraints, and operator workflow realities that cannot be replicated at RPI alone.

E. Pilot demonstration and robustness qualification under real conditions

In the final project phase, the system will be validated through pilot runs on SME-defined parts and

process settings, including changeover and re-calibration scenarios, repeatability checks, and stress tests (pose variation, payload/process variation). The final deliverable includes a deployment-ready package: software, calibration/verification procedures, test scripts, and a documented workflow that the SME (or an integrator) can adopt.

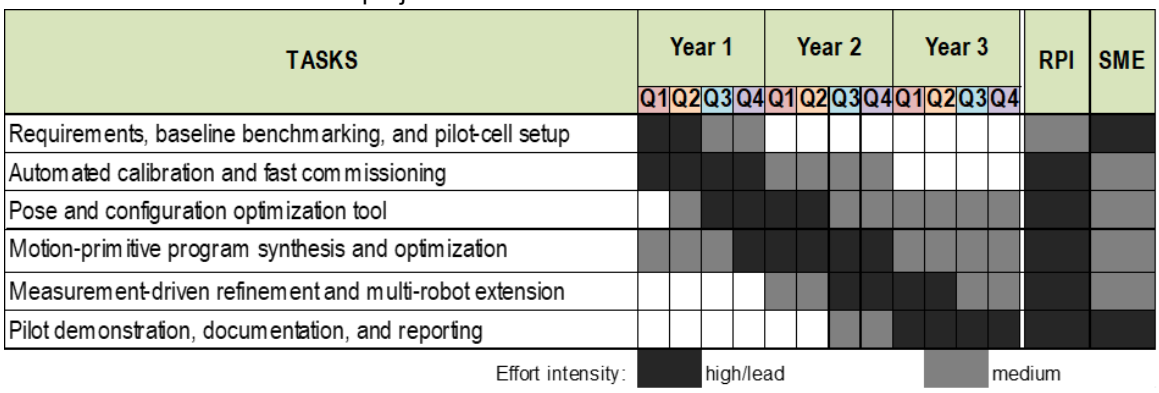
This overall approach ensures that SME-provided process knowledge and acceptance criteria drive the technical development, while RPI's core contributions in optimization, automated calibration, and controller-native program synthesis are implemented and field tested in a TRL 6 pilot demonstration.

3. Work Plan, Timeline, and Milestones

The project is planned for 36 months, with six tasks that are interleaved to ensure proper iterative development.

Task 1: Requirements, baseline benchmarking, and pilot-cell setup
Task 2: Automated calibration and fast commissioning
Task 3: Pose and configuration optimization tool
Task 4: Motion-primitive program synthesis and optimization
Task 5: Measurement-driven refinement and multi-robot extension
Task 6: Pilot demonstration, documentation, and reporting

The Gantt Chart of the overall project can be found below.



Milestones

Month 3: SME process & requirements definition completed
Month 6: Baseline benchmarking completed
Month 12: Demonstration of automated calibration and commissioning that meets specified requirement
Month 18: Demonstration of >15% cycle-time reduction versus baseline on controlled pilot tests while meeting accuracy/uniformity criteria across multiple representative paths.
Month 24: Demonstration of motion-primitive based program synthesis that meet accuracy and speed targets on representative curves with minimal manual tuning on at least two controller families.
Month 30: Demonstration on multi-robot or multi-part scenarios and robustness testing.
Month 36: Final demonstration, software and documentation transfer, submission of report.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI will lead the core algorithmic R&D and provide the scientific/engineering basis for performance, robustness, and portability. Specific roles include: 1. Algorithm development: constrained pose/configuration optimization; speed-maximizing planning under accuracy constraints; motion-primitive synthesis; controller-aware refinement and learning. 2. Calibration methodology: develop and validate automated calibration algorithms and drift monitoring. 3. Pilot experimental validation: design metrology and benchmarking procedures; analyze error sources; produce performance reports. 4. Research resources: provide research staff, software infrastructure, optimization/controls expertise, and testing protocols; host interim integration tests if needed. 5. Knowledge transfer: train SME engineers on workflows; deliver documentation and reference implementations.
Expected R&R of Korean SME Partner	The Korean SME partner will drive industrial relevance, integration, and TRL 6 validation through real-world constraints and pilot trials. Specific roles include: 1. Use cases and requirements: provide representative parts, process descriptions, quality criteria, and production constraints. 2. Pilot cell access: provide robot hardware/controllers, tooling, end-effectors, fixtures, and safety-compliant environment. 3. Integration engineering: support controller interfacing, offline/online program transfer, and shop-floor deployment workflows. 4. Testing and validation: execute production-representative trials; supply measurement data/inspection results; assess usability and maintainability. 5. Resource commitments: dedicate engineering time for joint testing, troubleshooting, and iteration; support procurement of sensors/metrology as needed. 6. Commercialization leadership: define productization pathway (internal adoption, integrator offering, licensing), customer discovery, and ROI metrics.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological achievements:** The project will deliver an SME-deployable, controller-ready software and procedures that unify automatic calibration/commissioning, pose/configuration optimization, and motion-primitive program synthesis to achieve high-speed, high-accuracy execution in single- and multi-robot manufacturing cells. We will demonstrate repeatable performance on SME-representative parts, targeting ≤4 hours initial commissioning (≤60 minutes re-calibration), ≤0.5 mm 3D RMS post-calibration accuracy, ≥25% cycle-time reduction versus current programs, and stable path-speed/contour performance enabled by a data-driven refinement loop that compensates controller- and cell-specific effects. **Commercialization potential:** The project will deliver a practical product package, consisting of software, procedures, and validation tests, that can be easily implement by SMEs or integrators to minimize engineering workload and decrease production downtime. Return on investment will be realized through faster changeovers, increased throughput, and reduced risk of scrap or rework. Likelihood of commercialization will be high due to wide applicability of the system. **Broader industrial/economic impact:** By making high-performance robotic manufacturing easier to deploy and maintain, the project will accelerate adoption of smart and convergent manufacturing in Korea, improve quality in robotic manufacturing processes (e.g., coating/cold spray, welding, grinding/polishing), and reduce reliance on scarce expert robot programmers. The

	resulting productivity gains and workforce upskilling will strengthen SME competitiveness and contribute to adaptable and higher-value manufacturing supply chains.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization